AGENDA ITEM 4 — STOCK ASSESSMENT THEME

4.1 Improvement of MULTIFAN-CL software

4.1.1. Update of MULTIFAN-CL software

171. There was no presentation, and SC21-SA-IP-02 (*Developments in the MULTIFAN-CL Software 2024-25*) was taken as read. There were no questions or comments raised from the floor.

4.1.2. Scoping the next generation of tuna stock assessment software (Project 123)

172. A. Magnusson (SPC) presented SC21-SA-WP-01 *Scoping the Next Generation of Tuna Stock Assessment Software: Progress Report, Project 123*, describing the current status of the scoping project launched in 2024 and the progress so far. The first part of the presentation was an evaluation of the necessary features for tuna assessment software. In addition to the technical features, three main criteria have been identified as requirements: scientific quality, beginner-friendly, and widely used. Stock Synthesis, Gadget, SBT, FIMS, and CASAL have been reviewed, and the recommendation is to launch a new project developing a new software platform in collaboration with IATTC and other tuna RFMOs. The second part of the presentation focused on two development work streams that have been initiated: DTU (Technical University of Denmark) spatio-temporal analysis of tagging data and a future SPC-IATTC model for tuna assessments. SPC organized a workshop in May 2025 with Tobias Mildenberger and Anders Nielsen at the DTU, starting a collaborative project that will analyze WCPO skipjack tagging data and aim to produce abundance indices that could then be incorporated into a stock assessment model. An online CAPAM workshop is planned for December 2025 as a collaboration between SPC and IATTC, focusing on the design of a new software platform for future tuna assessments. An initial concept outline has been proposed by Mark Maunder at IATTC, using the RTMB programming environment and starting with core features in the first development stage, adding later extensions in the second stage. A proposal will be presented at SC22 for the development project that will follow the scoping project. The current recommendation is to have the development project support both the DTU spatio-temporal tagging analysis and the development of a new SPC-IATTC assessment model, where the output from the tagging analysis could be used as data input in the assessment model.

Discussion

173. Tokelau, on behalf of FFA members, thanked the Project 123 team for their update in SC21-SA-WP-01. They congratulated them on the significant achievement of transitioning the two billfish stock assessments from Multifan-CL (MFCL) to Stock Synthesis. FFA members fully supported the paper's recommendations, particularly the convening of an intersessional working group to draft year-3 activities and TORs for Project 123. They welcomed the proposed collaborations with IATTC on developing a new tuna model based on RTMB, and the potential collaboration with the University of Denmark on analysing tagging data through an external spatio-temporal model, provided it remains focused on stock assessment development. FFA members further encouraged the team to strengthen partnerships with other tuna RFMOs, such as the CCSBT, which has rewritten its southern bluefin tuna stock assessment model over the last 6 months. They stated they would like to see work on actual code to progress as soon as practical, noting some urgency to work on the code for the next South Pacific albacore assessment in particular, because this will need to incorporate close-kin mark-recapture (CKMR) for the first time. They stated that these strategic collaborations are key to advancing robust, science-based fisheries management in the region.

174. The EU thanked the SSP for the overview, noting its complexity, and stated that the development of a new model seems to be the best option to provide a tailored solution that meets the needs of the stock assessments carried out in the WCPFC. They stated the approach clearly wins a high score in the first criterion for the selection of future assessment software, but asked for the SSP's views in relation to the two work streams proposed and the other two selection criteria (being user-friendly and having a large user community supporting it).

175. The SSP stated that this would be at the top of their minds in the coming months, and especially at the workshop in December to design the next tuna assessment software. They noted the inherent challenges of balancing high scientific quality, multiple features, model configuration options, and user friendliness. They stated that scientific quality is most critical, but it is also very important (perhaps more so at SPC than the other two RFMOs) that the model be usable by new scientists. They will seek to optimize the three criteria and maintain a focus on the auxiliary tools that help prepare the model input and tools that help in parsing, plotting, and diagnosing the model output. The SSP stated that working collaboratively would mean many people are involved in developing those tools, and they could be developed at the same rate as the main model, meaning the user interface would not be an afterthought. The SSP stated that ICCAT and IATTC are also very interested, which would provide access to expert help through technical advice and workshops, and those involved in non-tuna assessments globally may also be interested.

176. Chinese Taipei observed that some of the new model features are designed to fit a spatial-temporal framework for analysing tagging data and CKMR data and producing an abundance index. Traditionally, many of these indices have been estimated using logbook data provided by CCMs, which also contain valuable regional information, and inquired whether this new approach will replace the traditional CPUE standardization.

177. The SSP noted that these spatial-temporal models are quite different in their analytical approach from, for example, SPC's stock assessment models in that they deal with continuous space. So instead of having the tags assigned to some huge blocks of rectangular areas as we have today, they incorporate the fine locations. The DTU spatial-temporal model has many resemblances and similarities with SEAPODYM, for example, and has some resemblance to the spatial-temporal model that's used for CPUE. The SSP stated that these are three separate platforms: SEAPODYM addresses more ecological domain questions; the DTU model focuses on basically analyzing the tags to produce abundance indices, and can also produce movement coefficients between any arbitrary rectangular definitions; and CPUE analysis is separate, but the analytical equations used are not unrelated. They stated that it is an exciting thought whether they should be combined in some way, but at the moment, they are separate.

178. J. Hampton (SPC) observed that SEAPODYM is already a well-established spatiotemporal model that analyses tagging data, and SPC may already be a considerable distance along the way to be able to do the same sorts of things as is envisaged with the DTU team. He encouraged not discarding the idea of using SEAPODYM in that way, given that SEAPODYM already has a well-established framework for incorporating environmental data to drive movement estimation in a fairly realistic way, and can produce abundance indices, and estimates of absolute abundance if tags were analysed in more of a release-specific way, where tags would better inform fishing mortality and abundance. At the moment, the tags are used in a recapture condition mode that is mainly designed to inform movement estimation, but if the release condition mode for analysing the tags was incorporated, it would move closer to a spatiotemporal model, as envisaged with the DTU work.

179. The SSP agreed that SEAPODYM and the DTU spatial-temporal model have similarities in this sense, although SEAPODYM has limitations as a tool for producing abundance indices as it stands, because what is needed is both software and the team, and the SEAPODYM team is more than fully tasked with other research. The DTU spatiotemporal model may have some features that are very focused on this task of producing abundance indices that will make it a better tool, but the DNA of these models is strikingly similar, with minor features that are different. They stated the DTU spatial-temporal model is fully developed and was used in the 2024 EPO skipjack assessment to produce abundance indices. It will continue to be refined and improved, but it is presently operational, and SPC plans to secure funding to enable the analysis to be carried out.

180. The United States indicated it is fully committed to the success of Project 123 and is looking forward to working closely with the SSP and other CCMs at SC21. The USA stated it sees real value in aiming for a streamlined delivery timeline as outlined in the proposed schedule, with a functional core model in place within 2–3 years; this would allow benefits to be realized sooner, and enable them to be incorporated into upcoming assessments. The USA stated it has previously supported exploring collaboration with the NOAA Fisheries Integrated Modelling System (FIMS) development team, but their view now is that Project 123 may be better served by focusing efforts elsewhere, namely, continued collaboration with DTU on modelling the tagging data and development of a new model with IATTC. They stated that FIMS operates on its own pace and priorities, which may not align with WCPFC timelines. Concentrating on areas where tangible near-term progress can be made could yield more immediate benefits. As SC considers the next phase of Project 123 (which ends at SC22), the USA stated it expects that model development (including simulation testing) will likely entail a substantial workload. Recognizing the ever-increasing workload the SSP is asked to complete each year in support of existing Commission obligations, the USA suggested that when developing the proposal for phase 2, it includes plans to outsource the work from the SSP. In support of these model development efforts, the United States indicated it could contribute directly by supporting the development of a simulation testing framework, which could be an important step in evaluating new models, testing their performance under realistic scenarios, and ensuring the robustness of future stock assessments.

181. The SSP thanked the USA for its interest and support and for offering to assist with operating model development.

182. China stated that it is encouraged to see progress with the DTU and IATTC and considers this initiative to develop a more tuna-centric stock assessment model or software to be very important, acknowledging the collaboration between the SPC and other tuna RFMOs. China emphasized the importance of transparency, noting SPC had already developed a GitHub page to document all the progress over time, and encouraged this approach. China stated that the new coding, components, and even the simulation testing could be well documented to enable all CCMs to see the progress and have the opportunity to contribute. China also stressed wider engagement or involvement with different teams, observing that the workload will be huge to develop the new software, and stating that, in addition to scientists from the SSP and other tuna RFMOs, there may be people interested in contributing from the general scientific community. China observed it would be beneficial to broadcast this initiative more widely. China stated it is willing to contribute to the development of the new software, especially by contributing to different components of the RTMB coding, and looked forward to the collaboration.

183. The SSP concurred that the most successful way to develop open-source software today is to invite collaboration and stated that at the initial meeting (probably in December) they would establish a repository and encourage source code contributions and discussions on features.

184. Indonesia inquired whether, given that the tagging data is optional for Stock Synthesis, it is still necessary to collect tagging data for the future (2028) skipjack stock assessment, considering the substantial cost of the tagging program, and asked, in the context of the proposed new software, whether tagging data would still be required as input in the new software?

185. The SSP stated that the tagging data are extremely important for the skipjack assessment and that will always be the case. The modelling approach — whether the tags are analysed within the stock assessment model or externally and provided as input to the assessment model — will not affect the importance of tagging data, or the tagging protocols or resources. The results obtained over the next 12 to 18 months in terms of external analysis of tag data in a spatial-temporal model should indicate whether analysis external from the stock assessment model to produce abundance indices is feasible, or if a model that can internally analyse tags will still be needed.

186. Chinese Taipei encouraged the SSP to collaborate with outside contributors to do develop the software and inquired what criteria would be used in choosing which work stream to continue.

187. The SSP stated that the two work streams are not options in the sense that one would be chosen, but are two tools that are used together. The DTU spatiotemporal model is used for the important task of analysing the tags, whereas the stock assessment model, at least in its first version, would rely on the tags being analysed externally and provided as abundance indices, which will allow the stock assessment model to be much simpler than it would be otherwise. The SSP stated that the current conceptual design is that these tools will both be needed, and acknowledged that the development work would primarily take place outside of SPC. The DTU spatial model would be developed by the DTU team, and the model collaboration with IATTC would be headed by a part-time consultant for a few years.

188. The SSP stated while it will to be important to get the best available scientific information from the stock assessments, the small working group may consider the role of assessments as WCPFC moves into the harvest strategy approach, in which the stock assessments do not remain the major source of management advice on harvest levels, but are part of the monitoring strategy, making it important to consider that the role of assessments in the work of the scientific committee may evolve.

189. The theme convener observed that developing a model is a very specialized task that requires very detailed discussion on technical issues, but urged CCMs to keep in mind the big picture of the project.

190. China stated that this is an important project both for WCPFC and tuna management globally. China stated the project is important and necessary, and encouraged support for Project 123.

191. The United States supported Project 123 and thanked the SSP for taking into consideration their comments when it was first presented to focus on certain items that will provide a clear path for developing new software and strengthening collaboration with IATTC and other CCMs, starting with the workshop in December. The USA stated it is looking forward to working with the SSP and IATTC, and encouraged other CCMs to join the team.

Outcomes

192. SC21 thanked the SSP for their extensive work on Project 123, the Next Generation of Tuna Stock

Assessment Software, and acknowledged the project's progress, including the review of existing and ongoing software development projects.

193. SC21 noted that using a collaboration platform such as GitHub would facilitate continued collaboration with the wider scientific community in a consistent and transparent manner.

194. **SC21 recommended the SSP to collaborate with external contributors on this project to improve transparency, broaden expertise, and support the long-term development of the next-generation tuna stock assessment platform**.

195. The report from the Informal Small Group 07 (Project 123: Scoping the next generation of tuna stock assessment software) is included as **Attachment J.**

196. **ISG-07 provided the following project work areas to be conducted in 2026**:

- **DTU work on external tagging analysis**
- **Tuna model development collaboration (IATTC, etc); and**
- **Operational Model/simulation testing framework to evaluate model performance**.

197. **SC21 encouraged the continuation of work on the project with the revised 2026 work plan listed in the updated Project 123 Terms of Reference (TOR).**

4.2 Template for reporting stock assessment outcomes (Project 113b)

198. P. Neubauer (Dragonfly Data Science) introduced SC21-SA-IP-22 regarding changes that were asked for by the Commission (para 206, WCPFC21 Summary Report) and sought feedback on whether it meets the request by the Commission, which was recorded as:

- the inclusion of MSY-based reference points
- use an overfished reference status with respect to $SB/SB_{F=0}$
- revise the overfishing reference to F_{MSY}.

199. The intention was for the template to be modified for specific stock assessments, using the specific reference points and the specific management quantities that are accepted for specific stocks. It was also intended to reflect the assessment quantities and assessment advice as accepted by SC. Thus, once the assessment is presented and accepted, the assessment teams will include the assessment outcomes in the template, with the completed and hopefully accepted template and stock status table ultimately submitted to the Commission as part of the assessment advice.

Discussion

200. Australia, on behalf of FFA members, stated that on the template for reporting stock assessment outcomes, they noted the Commission's advice to include MSY-based reference points in the template 'if calculable and useful'. They stated that while MSY-based reference points are widely recognized and referred to in the WCPFC Convention as default benchmarks, caution should be used in their application, because MSY estimates are heavily influenced by poorly understood parameters in the stock–recruitment relationship and other assumptions that reduce their utility for prescription of management advice. FFA members stated that depletion-based RPs (such as $SB/SB_{F=0}$) are more appropriate benchmarks against which stock status should be reported when possible. They stated that, depending on the type of assessment, these may be calculated with greater confidence, are less sensitive to recruitment